

Wenxuan Wu
28010 stoney cloud drive
281-686-7086 | ww6726@tamu.edu
U.S. Citizen

EDUCATION

Texas A&M University <i>Doctor of Philosophy in Computer Science</i> GPA: 3.6 / 4.0; Advisor: Yupeng Zhang	College Station, TX/USA 09/2019- 08/2025
University of Texas at Austin <i>Bachelor of Science in Electrical Engineer</i> GPA: 3.5 / 4.0;	Austin, TX/USA 08/2014- 05/2019

SUMMARY

Result-driven cryptographic researcher specializing in public-key cryptography, applied zero-knowledge proofs, secure multi-party computation, and cloud computation. Demonstrated success in building and deploying end-to-end ZKP pipelines, new and efficient ZKP constructions based on various SNARKs.

TECHNICAL SKILLS

Programming Languages: C++, C, Python, Java, JavaScript, Rust, Circom, Solidity
Tools & Framework: Pytorch, AWS, Hadoop, Docker, Arkworks, Halo2, Databricks, Sparks, MP-SPDZ, Plonky, Openssl, Libsodium, Libsnark.

WORKING EXPERIENCE

TensorOpera AI <i>Research Engineer Intern, Web3</i> - Led ZKML (zero-knowledge machine learning) initiative, integrating zero-knowledge cryptographic techniques into ML workflows to enhance data privacy and security. - Engineered arithmetic circuits in Circom optimized for ZKP, applicable to large language models, advancing the frontier of privacy-preserving AI. - Transformed the federated ML network to its next generation by implementing token-based reward mechanisms, boosting agent participation, and enhancing the platform's competitiveness in attracting high-quality AI agents.	Sunnyvale, CA/USA 05/2023 – 08/2023
Retina Foundation of the Southwest <i>Software Engineer Intern</i> - Co-led the development of a CNN model to fully automate the retinal layer segmentation process, facilitating a shift from manual human grading to AI-assisted diagnostics, improving efficiency, and reducing diagnostic turnaround time. - Conducted iterative model improvements by optimizing image augmentation techniques to enhance model training and generalization, achieving comparable accuracy to human graders. - Collaborated with a multidisciplinary team of researchers and clinicians to gather requirements and validate model outputs for accuracy and reliability.	Dallas, TX/USA 05/2019 – 08/2019

PROJECT EXPERIENCE

Collaborative SNARK: Distributed Cloud ZKP Co-designed and implemented a new cloud-based, multi-prover zero-knowledge proof system (ZKP) using Halo2 and Rust.	05/2024 – 04/2025
---	-------------------

- Analyzed the pros and cons of different types of secret sharing techniques resulted in the team choosing packed secret sharing (PSS) as the baseline for our protocol's optimization
- Benchmarked our proof system. We deployed our protocol across 128 AWS EC2 instances and resulted in a 36x speedup compared to the state-of-the-art single prover ZKP, a 26x speedup compared to prior work.
- Maintained a succinct, memory-friendly ZKP. Our protocol preserved the memory advantage despite deploying it across 128 cloud instances.

RESCU: Verifiable ML on the Cloud

07/2022 – 09/2024

- Designed and implemented a delegated, and verifiable machine learning protocol in C++. This new protocol allows a client to delegate a ML inference task to an untrusted cloud, while maintaining the integrity of the inference as well as the privacy of client's data.
- Utilized secure multiparty computation (MPC) technique to perform all computations required for inference as well as verification.
- Benchmarked our protocol and tested it on MNIST dataset. Our protocol takes runs across AWS, Azure, and GCP instances and took 2.6 seconds to generate the inference on from a 3-layered CNN, 88x faster than the prior work.
- Collaborated with cross-functional teams from the U.S. Air Force Research Lab and MIT Lincoln Laboratory to ensure an efficient, up-to-date development environment and maintain project velocity.

Private Polynomial Commitment

12/2020 – 06/2022

- Designed and implemented a new cryptographic protocol, *Private Polynomial Commitment (PPCOM)*, improving privacy in polynomial verification by concealing both the evaluation point and polynomial details.
- Implemented a star-topology architecture for PPCOM using homomorphic encryption, allowing a central prover to securely execute polynomial evaluations and aggregations.
- Engineered a scalable multiparty private set intersection (MPSI) system on top of PPCOM, enabling secure collaboration among 1,000+ parties with zero added computation or communication cost.
- Implemented everything in C++. Optimized and benchmarked protocol performance, achieving 1,024 evaluations in under 3 minutes.

LEADERSHIP & ACTIVITIES

Activities: Teaching Assistant
Texas A&M University

College Station, TX/USA
09/2020 – 05/2021

- Mentored 200+ undergraduates during lab sessions, supporting assignment completion, midterm preparation.
- Assisted professor in preparing and delivering lectures, including creating instructional materials and demonstrations.